

OCEANIQ

Catching illegal oil spills at sea — and tracing them back to the ship responsible.

A plain-language walkthrough of the problem, our approach, and why it holds up against what already exists — written for anyone, technical or not.

01 — THE PROBLEM

Ships dump oil. Nobody can prove who.

Ships sometimes illegally discharge oil waste into the ocean instead of disposing of it properly, polluting the sea, harming marine life, and damaging coastal economies.

The hard part isn't spotting the oil — it's proving **which ship did it**. By the time anyone notices a spill, the ship has usually moved on. And vessels dumping illegally often deliberately switch off their location tracker to avoid getting caught.

The government (NTRO) wants a system that uses satellite images and ship-tracking data together to catch spills and point to the likely culprit.

02 — OUR IDEA

IN ONE SENTENCE

We work backward, like a detective: find the spill → trace it back through ocean currents and wind to where it most likely started → check which ships were actually there at that place and time → explain, step by step, why we suspect the ship we suspect.

03 — HOW IT WORKS

Seven steps, start to finish

- 1 Spot the spill using radar images**

Regular satellite photos need daylight and clear skies. We use radar satellite images instead, which see through clouds and darkness — exactly when illegal dumping is most likely.

- 2 Make sure it's actually oil**

Algae, calm water, and light rain can look like oil in radar images. A filter rules these out before we chase a false alarm.

- 3 Measure the spill**

Its size, shape, and direction become clues for the next step.

- 4 Trace it backward in time**

Using a trusted, existing ocean-science tool, we run the spill's drift **backward** — estimating roughly where and when it started, not just where it ended up.

- 5 Check which ships were actually there**

Ships constantly broadcast their location (like a GPS beacon). We pull historical ship data for that estimated time and place.

- 6 Score each nearby ship on multiple clues**

Right place? Right time? Consistent direction of travel? Tracker went quiet during the window? Every clue adds to a transparent score.

- 7 Show it all on a simple dashboard**

A map with the spill, the estimated origin area, and ship tracks — plus a ranked suspect list with a plain-language reason for each.

What already exists (being honest about it)

We're not claiming to invent something nobody has thought of:

- **CleanSeaNet**, run by the European Maritime Safety Agency, already does something similar for European waters.
- Researchers have published studies doing versions of this, including some over Indian waters.
- Some public tracking tools already flag ships that suspiciously go dark.

So what makes ours different?

WHAT OTHERS TYPICALLY DO	WHAT WE DO INSTEAD
Just detect the oil spill	Detect it and measure its shape and direction
Say "the nearest ship is guilty"	Trace the spill back to where it started, and check who was there then — not who's nearby now
Flag a missing tracker signal as proof of guilt	Treat it as one clue among several — trackers can fail innocently too
One vague confidence number	A clear breakdown of exactly which clues pointed to this ship, and how strongly

What we’re building — and what we’re skipping

BUILDING FOR THE DEMO

- Spill detection from a real satellite radar image
- Filtering out false alarms
- Backward drift tracing with an established tool
- Matching ship history to the estimated origin
- Explainable scoring and ranking
- A visual dashboard

SKIPPING FOR NOW

- Live, real-time coastline monitoring
- An automatic "100% guilty" verdict
- Anything needing expensive, non-public data

One example, start to finish

- 01 Satellite spots a spill
- 02 System confirms it's really oil, not a false alarm
- 03 System calculates where and when it likely started
- 04 System pulls up which ships were near that spot at that time
- 05 System ranks the ships and explains why the top one is the most likely suspect

08 — HONESTY

ONE LIMITATION, UPFRONT

We point to the **most likely** ship based on available evidence — we don't claim certainty. Think of it like a detective narrowing down suspects with strong evidence, not a court verdict. That's why we show our reasoning instead of hiding it behind one confidence number: it's meant to help human investigators focus, not replace them.